

# Week 3 Summary Report

AnalystLab Africa ML Internship

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## What I Learned

This week was about understanding how machine learning models are built and judged, not just how to train them. I covered the difference between supervised and unsupervised learning, why we split data into training and test sets, what overfitting and underfitting look like, and how to evaluate a model using accuracy and confusion matrices.

Supervised learning uses labeled data to predict an outcome. I used this on the Titanic dataset, where the target is whether a passenger survived. Unsupervised learning works with unlabeled data and looks for structure, like clustering similar passengers or samples together without being told the answer upfront.

I applied a train/test split on both the Iris and Titanic datasets to see how a model performs on data it has not seen before. This matters because a model can memorize training data and still fail badly in the real world. Testing on unseen data is the only honest way to know if it actually learned something useful.

On overfitting and underfitting: an overfit model performs great on training data but poorly on test data because it memorized noise instead of patterns. An underfit model performs poorly on both because it is too simple to capture the real relationship. I reduced these issues by tuning model complexity, using cross-validation, and scaling features properly.

For evaluation, I used accuracy and confusion matrices on both datasets. On Iris, my model got a perfect confusion matrix with zero misclassifications across all three classes. On Titanic, the confusion matrix showed a more realistic mix of correct and incorrect predictions, which helped me understand precision and recall in a way that accuracy alone does not show.

## Challenges Encountered

The main challenge was interpreting results correctly rather than just running code. Getting a perfect confusion matrix on Iris initially made me second guess myself, since 100 percent accuracy usually signals a leak or an easy dataset rather than a flawless model. I had to check my train/test split and confirm there was no data leakage before trusting the result.

I also found it easy to confuse precision and recall when explaining them in my own words, and had to work through examples on the Titanic data to really separate the two concepts instead of just repeating definitions.

## Key Insights

Accuracy alone can be misleading, especially on imbalanced data. A confusion matrix gives a clearer picture of where a model actually fails. I also learned that a dataset like Iris is a poor test of a model's real ability because the classes are so easy to separate, while Titanic is more useful for practicing real evaluation since it has more noise and imbalance.

Overall, this week shifted my focus from just building models to actually judging whether they are good enough to trust.